Project Planning & Management

Store Sales Dataset Analysis | Junior Data Analyst Project

Track	DEPI – Junior Data Analyst
Tool	Power BI
Deadline	16 Oct 2026
Team	Omar Mohsen El Sayed (Team Leader), Malak Nashat Abdalhamed, Raghda Yasser Mohamed, Asmaa Mohamed Mohamed
Supervisor	Dr. Ahmed Ayman
GitHub	https://github.com/Omar822006

1. Project Planning & Management

1.1 Project Proposal

Overview: This project analyzes the Superstore Sales dataset to help decision makers understand sales performance across product categories, regions, customer segments and time. The data is cleaned and modeled in Power BI, and the findings are presented in an interactive dashboard.

Dataset: Superstore Sales Dataset (Excel). Main columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales.

Objectives

- Clean and preprocess the dataset so it is ready for analysis.
- Define the business questions that matter to the organization's decision makers.
- Build a Power BI dashboard that answers those questions visually.
- Deliver a final report and presentation summarizing the work and insights.

Scope

- **In scope:** data cleaning, data modeling, analysis questions, dashboard design, final report and presentation.
- **Out of scope:** predictive modeling, real-time data integration, deployment to production systems.

1.2 Project Plan

Phase	Tasks	Tool	Deliverables	Due
Week 1 – Data Cleaning & Preprocessing	Clean and preprocess the data (fix date formats, handle duplicates, nulls and data types)	Power BI	Cleaned dataset, preprocessing notebook	Week 1
Week 2 – Analysis Questions	Define all analysis questions that decision makers would care about	Power BI	Set of analysis questions	Week 2
Week 3 – Dashboard	Build the dashboard that visualizes the answers	Power BI	Power BI dashboard	Week 3 (Implementation deadline: 30 Nov 2026)
Week 4 – Final Presentation	Prepare the report and presentation (analysis, insights, conclusions)	Power BI, PowerPoint	Final report and presentation	Week 4 (Final deadline: 4 Dec 2026)

Course milestones

Item	Deadline
Project Planning & Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis & Design	6 Nov 2026
Implementation (Source Code & Execution)	30 Nov 2026
Final Presentation, Testing & Reports	4 Dec 2026

All documents are uploaded to GitHub.

1.3 Task Assignment & Roles

Member	Role	Responsibilities
Omar Mohsen El Sayed	Team Leader / Project Manager	Planning, follow-up, GitHub repository, final report
Malak Nashat Abdalhamed	Data Cleaning Lead	Data preprocessing and quality checks in Power BI
Raghda Yasser Mohamed	Data Analyst	Analysis questions, DAX measures, insights
Asmaa Mohamed Mohamed	Dashboard Designer	Dashboard layout, visuals, formatting
All team members	Documentation & Presentation	Documentation, slides, final presentation

1.4 Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Data quality issues (inconsistent date formats, duplicates, missing values)	High	High	Systematic cleaning checklist; validate row counts before and after cleaning
Misunderstanding the dataset columns	Medium	Medium	Create a data dictionary early and review it with the team
Missing deadlines	Medium	High	Weekly check-ins; split tasks with clear owners
Team member unavailable	Low	Medium	Share files on GitHub; cross-train members on key tasks
Dashboard too cluttered or unclear	Medium	Medium	Review design with the lecturer; keep one question per visual
Loss of work or version conflicts	Low	High	Commit regularly to GitHub; keep clear file naming

1.5 KPIs (Key Performance Indicators)

Project success KPIs

- 100% of deliverables submitted on or before each deadline.
- Data quality: 0 duplicate rows and 0 invalid dates after cleaning.
- All defined analysis questions answered by at least one visual.
- Dashboard refreshes without errors and loads quickly.
- Positive feedback from the lecturer on clarity and insights.

Business KPIs shown in the dashboard

- Total Sales
- Sales by Category / Sub-Category

- Sales by Region / State
- Sales by Customer Segment
- Sales trend over time (monthly / yearly)
- Average shipping time (Ship Date – Order Date)
- Top 10 products and customers by sales